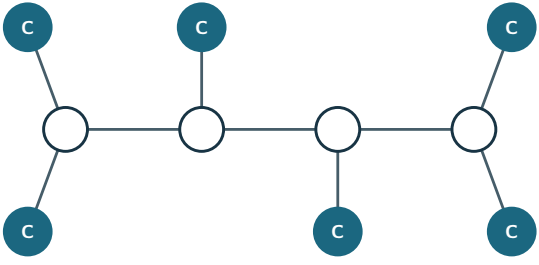
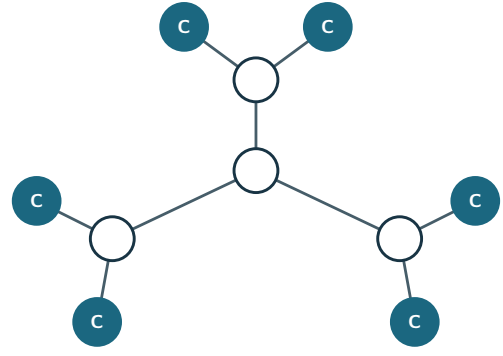
# Trivalent regions between circle operators

The number of bare junctions may be any fixed finite number.

**A path of four bare junctions**




**A branching bare region**




bare product junction

C vertex with a crossing-circle operator

Every bare junction has degree three, including its boundary attachments.
Every edge is a physical quarter-turn. Circle-marked vertices may have further branches.

The graphs record the original tangle operations.
Bare vertices are the product junctions after consecutive products have been grouped.